Final Project Evaluation

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Problem Statement

Input:

A csv file containing the columns -
Date, Open, High, Low, Close, Adj Close,Volume

Output:

Predicted closing prices for the given stock over a future time window.

Example:

Input: `1962-01-15,7.214786,7.237094,7.214786,7.221160,1.576214,266730`

Output: `7.049076`

The goal is to model and predict future closing prices using sequential data. Initially, a Hidden Markov Model (HMM) was used, and later enhanced using GMM-HMM-LSTM for capturing both temporal patterns and hidden market states .

Motivation

- Stock price prediction is a crucial challenge in quantitative finance.
- Markets are influenced by multiple stochastic and temporal patterns, which makes modeling difficult but impactful.
- The project aims to leverage sequence modeling techniques to capture latent market states and predict prices more reliably.
- Real-world application: aiding traders and investors in decision-making by forecasting short-term trends

Literature Review

- Various approaches have been explored in the quest for reliable prediction models as accurate forecasting of stock prices and identification of trends are crucial for investors and financial institutions seeking informed decisions.
- Before the emergence of artificial intelligence, probabilistic techniques formed the foundation of stock market forecasting. These theories encompassed random walk models, correlation-based methods, scaling properties, stock market volatility and investor sentiment, probability distributions of stock price returns, and other relevant approaches.

Literature Review continued…

- Support Vector Machines were quickly recognized as promising for time series prediction, and were first employed for financial forecasting in 2001. Hybrid models combining ARIMA and SVM were introduced in 2005.
- Neural networks, including specific types coupled with Genetic Algorithms, have been utilized to detect temporal patterns
- **HMM and HMM-LSTM were used** in our approach because HMM captures latent market regimes and temporal dependencies effectively, while LSTM enhances this by modeling longer-term sequential patterns and non-linear relationships in stock prices, leading to more accurate predictions

Dataset

Example Data Instance:

- ```
  Date,Open,High,Low,Close,Adj Close,Volume
  1962-01-02,7.374124,7.374124,7.291268,7.291268,1.591517,407940
  ```

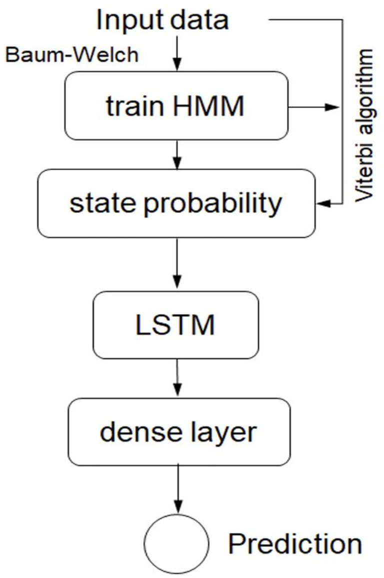
Statistics:

- Samples : ~15,000 Data rows (60 years of stock prices of a company
- Features: 5 numeric columns (excluding date)
- Dataset source : HMM-Stock-Market-Prediction/datasets/csv/AAPL.csv at main · valentinomario/HMM-Stock-Market-Prediction · GitHub
  ```

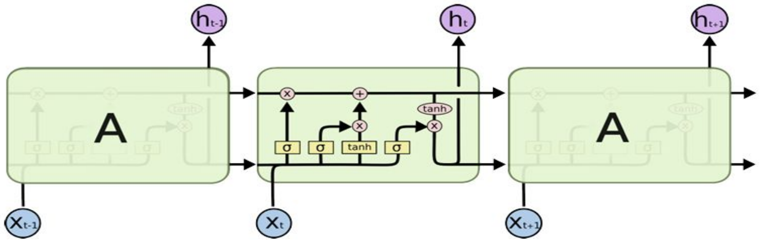
Method/Technique

- The HMM captured hidden market regimes (e.g., bull, bear) by modeling probabilistic transitions using features like log return, volatility, and volume change. The output hidden states and their state-wise means were appended as new features, enriching the data passed to the LSTM.
- The LSTM learns temporal dependencies and long-term patterns in the stock price sequences, using both original stock data and HMM-derived features to improve prediction of future closing prices

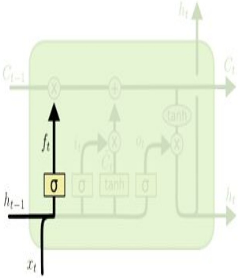
Method/Technique continued…

- **LSTM Architecture**: Consists of two stacked LSTM layers (50 units each) followed by dense layers; trained using Mean Squared Error (MSE) loss and the Adam optimizer. Inputs were normalized using MinMaxScaler for stability.
- 

Method/Technique continued…

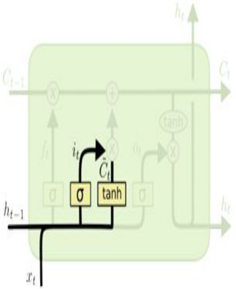


LSTM Process



$$f_t = \sigma\left(W_f\cdot[h_{t-1}, x_t] + b_f\right)$$

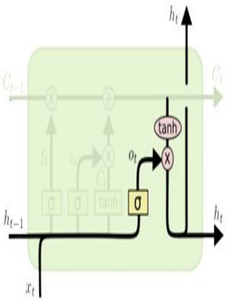
Forget Gate Layer



$$i_t = \sigma\left(W_i\cdot[h_{t-1}, x_t] + b_i\right)$$

$$\tilde{C}_t = \tanh(W_C\cdot[h_{t-1}, x_t] + b_C)$$

Input Gate Layer



$$o_t = \sigma\left(W_o\,[h_{t-1}, x_t] + b_o\right)$$

$$h_t = o_t * \tanh\left(C_t\right)$$

Output Gate Layer

Method/Technique continued…

- Implementation Details: Used libraries like hmmlearn, sklearn, keras, numpy, and pandas. HMM was configured with 3 hidden states using GaussianHMM. LSTM was trained for 25 epochs with batch size 64, using the Adam optimizer and Mean Squared Error (MSE) loss. Input features were scaled using MinMaxScaler, and the model used a 30-day lookback window for sequence input

Results

- Mean Average Percentage Error (MAPE)
  HMM - 1.1 %
  HMM LSTM - 1.1 %
  ARIMA - 1.8%
- Directional Prediction Accuracy (DPA)
  HMM - 41%
  HMM LSTM - 60%

Analysis

- The hybrid model better captures volatility and extreme price changes, and has a better direction prediction accuracy.
- LSTM improves short-term forecasting by learning memory of price swings.
- GMM-HMM provides interpretable market states, useful for financial insights.

Error analysis

- Although the DPA is better for the combined HMM-LSTM model than HMM, there is still considerable misclassification on the direction of stock movement within a day.
- Stock market movements are influenced by **unpredictable external events** (e.g., political news, global crises), which our model cannot account for with historical price data alone.
- LSTM models struggle with **short-term fluctuations** or abrupt market reversals that are not reflected in the prior window of input data.

Improvements over the paper

- Combined GMM-HMM with LSTM instead of using them separately.
- Enhanced temporal modeling by feeding latent states to LSTM instead of raw features.
- Further improvements which can be implemented -
  - Better Feature Engineering and Parameter Tuning
  - Extend LSTM to output prediction intervals instead of point estimates

Learnings

- We gained hands-on experience with Recurrent Neural Networks, especially LSTM architectures, learning how they handle sequential dependencies in time-series data.
- Learned the importance of deriving meaningful features like log returns, volatility, and how HMM-derived hidden states can enhance model input.
- Improved our skills in collaborative research, iterative testing, and interpreting real-world financial data

Demo

Summary and Conclusion

- **Problem:** Predicting stock closing prices using sequential models.
- **Approach:** Started with HMM → upgraded to GMM-HMM-LSTM hybrid.
- **Outcome:** Improved Directional prediction accuracy and speed
- **Future Work:** Incorporate external features like sentiment, news, macroeconomic indicators.